POLITECNICO
MILANO 1863

Advanced Computer Architectures

June 30, 2017

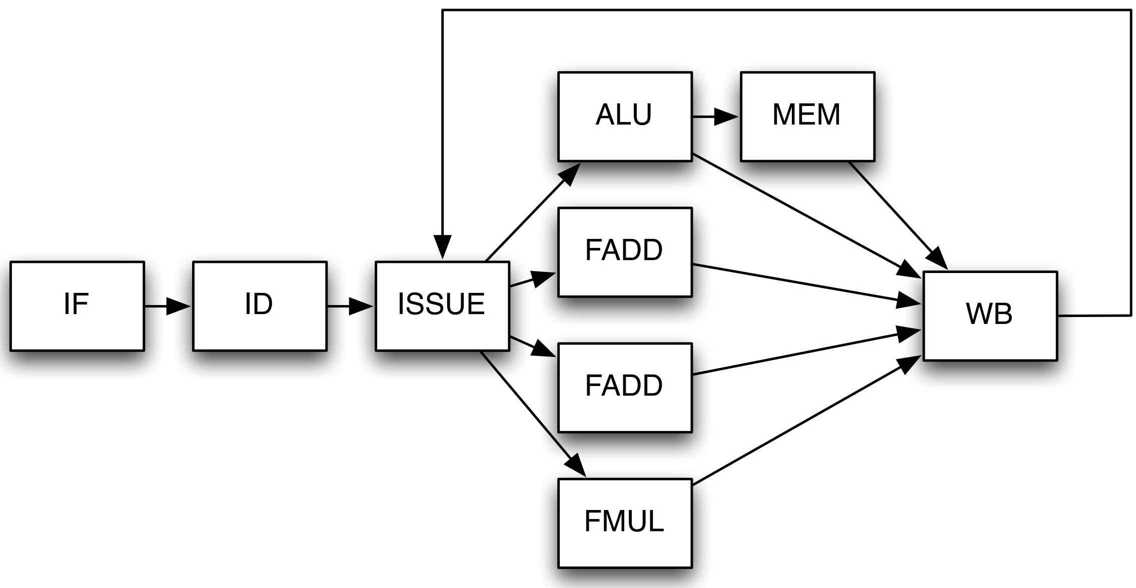
Prof. Donatella Sciuto

Name	
Last Name	
POLIMI ID Number	

Problem 1 (6 points)	
Problem 2 (5 points)	
Problem 3 (5 points)	
Question 1 (5 points)	
Question 2 (6 points)	
Question 3 (5 points)	
Total (32 points)	

Problem 1

In this problem we will examine the execution of a code segment on the following single-issue out-of-order processor:



You can assume that:

- All functional units are pipelined
- ALU operations take 1 cycle
- Memory operations take 3 cycles (includes time in ALU)
- The Write Back unit has a single write port
- There is no register renaming
- Instructions are fetched, decoded and issued in order
- An instruction will only enter the ISSUE stage if it does not cause a WAR or WAW hazard
- Only one instruction can be issued at a time, and in the case multiple instructions are ready, the oldest one will go first

Fill in the following table to indicate how the given code will execute. Assume all register values are available at the start of execution.

INSTRUCTION	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	Type of Hazards
`lw $t1,VECTA($t6)`															
`addi $t1,$t1,4`															
`sw $t1,VECTA($t6)`															
`lw $t1,VECTB($t6)`															
`addi $t1,$t1,4`															
`sw $t1,VECTB($t6)`															

Insert in the empty pipeline scheme the stalls needed to solve the previous data, control and structural hazards.

INSTRUCTION	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	C15	C16	C17	C18	C19	C20	C21	C22	C23	C24
`lw $t1,VECTA($t6)`																								
`addi $t1,$t1,4`																								
`sw $t1,VECTA($t6)`																								
`lw $t1,VECTB($t6)`																								
`addi $t1,$t1,4`																								
`sw $t1,VECTB($t6)`																								

Fill in the following table pointing out the NUMBER OF STALLS to be inserted before (or during) each instruction to solve data, control or structural hazards.

INSTRUCTION	Type of Hazards	Number of stalls
`lw $t1,VECTA($t6)`		
`addi $t1,$t1,4`		
`sw $t1,VECTA($t6)`		
`lw $t1,VECTB($t6)`		
`addi $t1,$t1,4`		
`sw $t1,VECTB($t6)`		

Problem 2

Please consider the program in the table be executed on a CPU with dynamic scheduling based on TOMASULO algorithm with:

- 2 RESERVATION STATIONS (RS1, RS2) + 2 LOAD/STORE unit (LDU1, LDU2) with latency 4
- 2 RESERVATION STATIONS (RS3, RS4) + 2 ALU/BR FUs (ALU1, ALU2) with latency 2
- We assume 1 CDB for RF, values on the CDB can be used by the units in the next clock cycle
- Static Branch Prediction ALWAYS TAKEN with Branch Target Buffer

1. Please complete the TOMASULO TABLE by assuming all cache HITS and considering **ONE ITERATION**:

ISTRUZIONE	Prediction T /NT	ISSUE	START EXEC	WRITE RESULT	Hazards Type	RSi	UNIT	OP1	OP2	STORE BUFFER
FOR:lw $t1,VECTA($t6)										
addi $t1,$t1,4										
sw $t1,VECTA($t6)										
lw $t1,VECTB($t6)										
addi $t1,$t1,4										
Sw $t1,VECTB($t6)										
addi $t6,$t6,4										
blt $t6,$t7, FOR	T									

2. Express the **formula** then calculate the following metrics:

CPI = ______________________________

IPC = ______________________________

Problem 3

Design and describe (all the decisions have to be motivated) a 1-BHT and a 2-BHT able to execute the following assembly code.
(R0 is set to 1, R1 is set to 0)

```
LOOP:     LD F3 0 R0
          ADDD F1 F3 F3
          MULTD F2 F3 F1
          ADDI R1 R1 10
LOOP2:    MULTD F2 F2 F3
          SUBI R1 R1 1
          BNEZ R1 LOOP2
          SUBI R0 R0 1
          BNE R0 R1 LOOP
```

How many times the BNEZ instruction is executed?
How many times the BNE instruction is executed?

Calculate the percentage of mispredictions for 1-BHT and 2-BHT by considering a generic table size.

Question 1

Complete the following table, indicating for each type of exception their characteristics with respect to the given columns.

Exception type	Synchronous/ Asynchronous	User requested/ coerced	Within or between instructions
Hardware malfunction			
I/O request			
OS call			
Arithmetic overflow			
Memory illegal access			

1. Define the concept of precise exception.
2. Explain why exceptions can occur out of order in an in-order, 5-stage pipeline and describe how such a pipeline reorders these exceptions to provide a precise exception model.

Question 2

Describe the architecture of **vector processors.** Explain the advantages of vector processing with respect to scalar, by explaining also the concept of vector chaining.

Question 3

Why a second or third level cache is useful?
What kind of benefits it provides to the system performance?
Explain how it impacts by showing the Average Memory Access time formula.
Name one other technique of cache improvement that can provide a solution to the same problem